

Video2Track: From Real-World Interaction Videos to Steerable Adversarial Closed-Track Testing for Automated Driving Systems

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ABSTRACT

Closed-track testing plays a fundamental role in the verification and validation of automated driving systems (ADS), particularly for safety-critical scenarios, by enabling reproducible evaluation under controlled conditions. However, most existing approaches still rely on standardized protocols or predefined trajectories, leading to overly scripted interactions and limited ability to reproduce the natural complexity of public-road traffic. To address this limitation, we propose Video2Track, a framework that transfers real-world interactive driving scenarios from videos into steerable adversarial closed-track testing. The framework consists of two tightly coupled modules. The first is a scenario semantic mapping module, which extracts structured semantics from driving videos using a vision-language model and grounds them onto a closed-track topology library via retrieval-augmented generation, thereby identifying compatible map segments and interaction anchors. The second is a dynamic interactive testing module, which conditions on the grounded topology and anchors to generate diverse multi-agent trajectories through a conditional diffusion model, while regulating interaction intensity via a Stackelberg game with a parameterized adversarial objective. Closed-track experiments demonstrate that the proposed framework can faithfully reproduce representative real-world interaction scenarios and generate executable scenario variants with controllable risk levels and interaction styles, providing a scalable approach for realistic and steerable ADS validation.

1. Introduction

Automated driving systems (ADS) operate in highly interactive, dynamic, and uncertain traffic environments, making comprehensive safety validation essential for real-world deployment [1, 2, 3]. Compared with simulation, real-world testing more faithfully captures ADS performance under real dynamics, actuator constraints, and multi-agent interactions, while closed-track testing, as an important real-world testing paradigm, provides a safe, controllable, and repeatable environment for ADS validation. However, existing closed-track testing still largely relies on standardized protocols, manual configuration, or predefined trajectories [4], with background vehicle (BV) behaviors heavily scripted and interactions relatively simplistic, thereby limiting the ability of such tests to faithfully reproduce the complex and evolving interactions observed on public roads. Although closed-track testing offers high controllability, this lack of realism may prevent its outcomes from fully reflecting the actual on-road performance of ADS, which in turn explains why public-road testing remains widely relied upon in both industry and research.

Although public-road testing exposes ADS to the most realistic traffic flows and interaction behaviors, it also involves high safety risks, low exposure frequency of safety-critical events [5, 6, 7], and limited scenario reproducibility, making it difficult to support systematic and efficient safety

validation. Meanwhile, despite recent advances in closed-track testing facilities and tools, such as virtual-real integration [8, 9] and cloud-controlled testing systems [10, 11], a critical gap remains in effectively transferring natural public-road interactions into closed-track testing. This raises a fundamental question: how can such interactions be faithfully reproduced on closed tracks for specific ADS testing purposes? Here, “fidelity” does not refer to simply replicating the original trajectories, but to preserving the essential road topology, interaction structure, conflict evolution, and risk characteristics of the source scenario while transforming it into an executable, steerable, and repeatable test case. Solving this problem is key to moving closed-track testing beyond scripted validation toward realistic interaction reproduction, thereby enabling safety evaluation with both realism and engineering practicality.

Many studies have explored incorporating real-world traffic data into ADS testing through mechanism-based modeling [10, 12, 13] and data-driven approaches [14, 15]. While these methods improve scenario realism, diversity, and adversariality, they primarily reproduce statistical patterns or risk characteristics, with limited ability to preserve consistency with specific real-world events in terms of interaction structures and temporal evolution. With the advancement of large models and diffusion-based approaches [16, 17, 18], increasing attention has been devoted to generating test scenarios from natural driving videos, which are widely available from onboard sensors such as front-facing cameras and dashcams. By leveraging vision-language models (VLM) to extract structured scene semantics from videos, these methods provide more expressive semantic representations

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than trajectory-based approaches. However, they are still primarily developed and evaluated within simulation environments, where generated scenarios are typically predefined or script-driven and lack closed-loop interaction with the vehicle under test (VUT), thereby limiting their applicability in topology-constrained closed-track testing environments that require strong interaction.

This gap gives rise to a fundamental challenge: how to transfer video-derived scenarios into executable closed-track testing cases. Driving videos capture rich interaction processes among traffic participants under specific contextual conditions; however, the extracted semantics must be grounded in the topology of closed-track testing environments. Moreover, the generated trajectories should not only preserve the original interaction logic, but also remain kinematically feasible and support controllable risk variation. Accordingly, a video-driven scenario transfer framework should satisfy three key requirements: (i) *fidelity*, which ensures the preservation of interaction semantics, participant relationships, critical conflict structures, and risk characteristics observed in the source video; (ii) *adversariality*, which enables continuous regulation of interaction intensity and risk exposure while maintaining the underlying interaction structure, thereby supporting stress testing; and (iii) *steerability*, which allows the generation of controllable scenario variants with adjustable risk levels and interaction styles.

To address these challenges, we propose Video2Track, a framework that migrates real-world driving interactions from videos into steerable adversarial closed-track testing scenarios. Specifically, Video2Track first extracts structured scenario semantics from raw driving videos and aligns them with a closed-track topology library, thereby grounding interaction events onto executable test environments. Conditioned on the grounded map constraints and interaction anchors, the framework then generates dynamically feasible test trajectories and further regulates interaction intensity to enable controllable adversarial evaluation. The main contributions of this work are summarized as follows:

- (1) We propose Video2Track, a framework for transferring real-world scenarios from interaction videos into closed-track testing, thereby overcoming the scripted nature of conventional closed-track tests and enabling faithful reproduction of natural public-road interactions.
- (2) We develop a retrieval-augmented scenario semantic mapping method that bridges VLM-extracted interaction semantics and deployable closed-track topology, enabling executable scenario construction on closed tracks.
- (3) We develop a dynamic interaction testing method that combines interaction-guided diffusion trajectory generation with Stackelberg-based interaction risk regulation, enabling steerable adversarial testing with controllable interaction risk and interaction style.

The remainder of this paper is organized as follows. Section 2 reviews related work; Section 3 presents the Video2Track framework and its key modules; Section 4

reports the closed-track experimental results and analysis; and Section 5 concludes the paper and outlines directions for future work.

2. Related work

2.1. Scenario Generation Methods

Conventional scenario generation methods can be broadly categorized into rule- and knowledge-based approaches and data-driven approaches. Rule- and knowledge-based methods construct scenarios from expert knowledge, standard protocols, or predefined templates, and instantiate them through constraint solving [10], N -wise coverage [19], or parameter sampling [20]. Representative frameworks such as PEGASUS [21] and Scenic provide structured scenario modeling and probabilistic specification under explicit geometric and semantic constraints, offering strong interpretability, controllability, and reproducibility. However, their reliance on manual abstraction and template design often leads to fixed interaction structures and scripted behaviors, limiting their ability to capture the natural variability of real-world driving interactions. In contrast, data-driven methods learn traffic distributions and interaction patterns from large-scale naturalistic driving data, thereby improving scenario realism and diversity [22]. Representative works include TrafficGen [23], which generates vehicle layouts and long-horizon multi-agent trajectories through an autoregressive model, CTG [24], which improves controllability via STL-guided diffusion, and DiffScene [16], which models the joint distribution of multi-agent motion using conditional diffusion. Although these methods improve diversity and distributional realism, they do not necessarily preserve the high-level interaction semantics required by specific testing objectives, nor do they explicitly address deployment requirements such as closed-track topology compatibility and vehicle dynamics feasibility.

More recent studies have begun to integrate VLMs, LLMs, and generative models to bridge visual-language understanding and scenario generation. The central idea is to extract structured scene semantics from images or driving videos and then translate them into executable scenario scripts or conditioning signals for downstream generation. For example, ScriptGPT [25] converts dashcam videos into Scenic scripts for real-to-sim reconstruction in CARLA; OmniDrive [17] combines multimodal perception and simulation platforms to generate realistic and diverse traffic scenarios for simulation-based testing and evaluation of AV; DriveGen [18] uses VLM-extracted waypoints to guide diffusion-based trajectory generation for ADS testing; and [26] translates driving videos into executable Scenic code for simulation-based safety testing in CARLA. While these methods improve scenario understanding and provide more intuitive interfaces for scenario specification, most remain simulation-oriented, where visual semantics are translated into Scenic scripts, OpenSCENARIO representations, or auxiliary constraints rather than being consistently grounded in deployable closed-track topology and executable vehicle behaviors.

2.2. Scenario-based Adversarial Testing for ADS

Scenario-based testing based on standardized protocols is widely adopted in industry as well as by regulatory organizations [27]. Such approaches explicitly define scenario configurations, target objects, speed profiles, overlap ratios, and evaluation criteria, thereby ensuring strong reproducibility and comparability. For example, in AEB testing, Euro NCAP specifies a variety of vehicle-to-vehicle collision configurations, such as rear-end and crossing-path scenarios, along with detailed parameter settings for each case [28]. In addition, the ISO 34502 [29] scenario-based safety evaluation framework provides systematic guidance for ADS safety assessment workflows, further reinforcing the methodology of organizing verification and validation around structured scenarios. However, such standardized protocols are inherently limited to predefined scenario configurations and lack the ability to represent dynamic and adaptive interactions among traffic participants.

Recent studies have explored search-based and adversarial testing approaches to generate more challenging interaction scenarios for ADS evaluation. The core idea is to search over scenario parameter spaces or environmental perturbation spaces to identify cases that trigger system failures or lead to safety-critical conditions, as measured by metrics such as time-to-collision (TTC), post-encroachment time (PET), and minimum distance. A representative method is worst-case scenario evaluation (WCSE) [30], which seeks the most challenging test cases by searching for worst-case conditions. While effective in exposing system vulnerabilities, such approaches may overemphasize extreme events and produce unrealistic scenarios that deviate from natural traffic distributions. Reinforcement learning (RL)-based approaches [31, 32] have also been explored for adversarial testing. A representative example is D2RL, which densifies safety-critical information and trains BV policies in simulation environments (e.g., SUMO) to accelerate the discovery of failure cases. Although effective for exposing rare but critical events, these methods rely heavily on simulation-based learning and are subject to distribution mismatch between simulation and real-world traffic, which may limit their realism and applicability to real-world testing.

Another important direction is game-theoretic testing, which explicitly models the mutual dependence among interacting agents and is therefore well suited for interactive scenario modeling. Level- k game frameworks construct differentiated interaction strategies under varying driver styles, providing a structured way to evaluate ADS behavior in multi-agent interactions [33]. More recently, Trouble-Maker [10] represents a state-of-the-art closed-track testing framework, where the ADS and BVs are modeled as leader-follower agents along predefined reference paths to generate diverse adversarial interactions. However, game-theoretic approaches typically depend on predefined reference trajectories or manually designed interaction structures, which limits their ability to generate diverse scenarios and faithfully capture real-world interaction semantics. Consequently, existing methods still lack a unified framework

for semantically consistent, executable, and controllable adversarial testing in closed-track environments.

3. Methodology

This section presents the technical details of Video2Track, the proposed framework for migrating scenarios from real-world interaction videos to steerable adversarial closed-track testing for ADS.

3.1. Framework Overview

As illustrated in Fig. 1, the proposed framework consists of two main components: scenario semantic mapping and dynamic interactive testing. First, real-world interaction videos are used to extract high-level scenario semantics through a VLM. These semantics are then grounded onto the proving-ground topology library to identify the target map topology and interaction anchors. Based on the grounded semantics, an interaction-guided diffusion trajectory generator produces naturalistic and dynamically feasible multi-agent trajectories as behavioral priors. A Stackelberg-based interaction risk regulation module then further adjusts the interaction risk based on these priors, thereby enabling steerable adversarial testing.

In the scenario semantic mapping stage, interaction videos, together with a scenario prompt, are processed by a VLM to extract structured scenario semantics, including road topology, agent states, and interaction event. These semantic representations capture the essential interaction structure of real-world scenarios. To instantiate the scenario in a closed-track environment, the extracted semantics are mapped to a topology library constructed from OpenDRIVE maps. Leveraging a retrieval-augmented generation (RAG) mechanism [34], the system retrieves the most suitable topology segment and instantiates corresponding event anchors, including initial states, goal states, and conflict points.

In the dynamic interactive testing stage, the retrieved topology and anchor constraints guide a diffusion-based multi-agent trajectory generation module to generate feasible interaction trajectories. To enable controllable risk exposure, an interaction regulation module formulated as a Stackelberg game is introduced, where a risk parameter modulates the interaction intensity by adjusting trajectory optimization objectives. The resulting trajectories can be executed in real-world closed-track experiments, enabling controllable and reproducible evaluation of autonomous driving systems.

3.2. Scenario Semantic Mapping

3.2.1. Scenario Abstraction

In Video2Track, real-world driving videos are first abstracted into structured high-level interaction semantics using a vision-language model (VLM). In the present framework, this module is instantiated with OpenAI GPT-5.4. For this stage, the *VideoScenario* dataset [35], a publicly available dataset collected from typical interactive conflict scenarios, is used. This stage aims to extract structured,

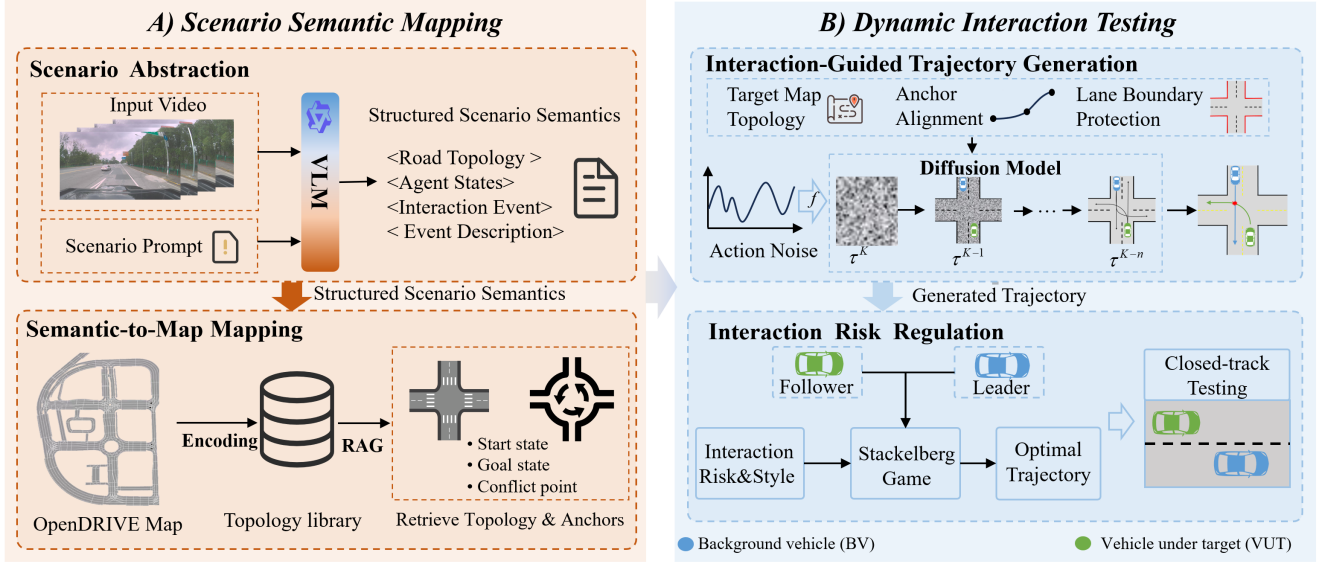


Figure 1: Overview of the proposed Video2Track framework. It consists of two main components: scenario semantic mapping, which combines VLM-based semantic extraction with topology-library retrieval for semantic grounding, and dynamic interactive testing, which integrates diffusion-based multi-agent trajectory generation with game-theoretic interaction risk regulation for steerable closed-track execution.

interaction-relevant semantics from videos, which are subsequently used as semantic constraints for semantic-to-map grounding.

A well-designed prompt is critical for improving the reasoning efficiency of large models [36]. We adopt a structured Chain-of-Thought (CoT) prompting strategy for the VLM, where reasoning proceeds hierarchically from road topology to ego behavior and then to interacting agents. Specifically, the model first identifies the road type and lane structure, then infers the ego vehicle's lane and maneuver intention, and subsequently detects the main interacting BVs and their behaviors. Based on trajectory relationships and observed behavior patterns, the interaction event, interaction type, and interaction style are further inferred. The extracted semantics are represented in a structured form, including road topology, agent states, interaction events, and an event-level description, which serves as an intermediate representation for downstream scenario grounding and trajectory generation. Details are provided in Appendix A.

Under the above CoT reasoning mechanism, suppose the input video consists of T frames:

$$V = \{I_t\}_{t=1}^T \quad (1)$$

where T denotes the number of frames and I_t is the t -th frame.

Given the scenario prompt P , the VLM maps the input video V to a structured scenario semantic representation:

$$s = F(V, P) = \left(\Psi, e, \{b_i, \phi_i, \gamma_i\}_{i=1}^N, M \right) \quad (2)$$

where Ψ denotes the road topology, including the road type and lane structure; e denotes the ego-vehicle state, including its initial position within the topology and maneuver intention; each b_i denotes the state of an interacting BV, including

its initial relative position and driving behavior; ϕ_i denotes the interaction type between the ego vehicle and the i -th BV (e.g., unprotected left-turn or cut-in); γ_i denotes the interaction style (e.g., rushing or yielding); and M is an event-level natural-language description of the overall interaction scenario. Here, N denotes the number of interacting BVs, which is typically no more than two. The risk parameter, quantified by post-encroachment time (PET), is not inferred by the VLM but directly obtained from the label associated with each source video.

3.2.2. Semantic-to-Map Mapping

The extracted interaction semantics are grounded onto the topology of the target closed-track map, thereby bridging the semantic and geometric layers of scenario representation. To support this process, we construct a topology library by parsing the closed-track map represented in the standard OpenDRIVE format. The library consists of representative road structures, such as intersections, straight segments, and merging segments. Each topology fragment is annotated with executable anchors, including entry anchors, exit anchors, and lane identifiers, all associated with physical coordinates on the site. These anchors enable executable testing scenarios to be instantiated by constraining the mapping from semantic descriptions to geometric realizations.

Given the road topology description Ψ in the VLM-derived semantic representation, a semantically consistent topology fragment is retrieved from the topology library through a RAG mechanism [37]:

$$\mathcal{M}^* = \text{Retrieve}(\Psi, \mathcal{D}) \quad (3)$$

where $\mathcal{D}$ denotes the topology library and $\mathcal{M}^*$ represents the retrieved target topology fragment.

Once the topology fragment is retrieved, semantic-to-map grounding is performed in a step-wise manner. The ego vehicle is first instantiated by mapping its start region to an entry anchor and selecting a goal anchor according to its maneuver intention, followed by route generation on the lane-level graph. Interacting BVs are then instantiated based on their relative spatial relations and interaction types, with corresponding entry and exit anchors selected to derive feasible routes. The conflict anchor is identified by matching the semantic interaction description with the pre-annotated anchors in the topology library.

Through this process, semantic descriptions are grounded into geometric constraints, including start states, goal states, and conflict anchors, enabling closed-track testing. It is worth noting that this stage only determines geometric anchors and topological constraints, rather than generating concrete trajectories. Details of the encoded topology library are provided in Appendix B.

3.3. Steerable Adversarial Testing

3.3.1. Interaction-guided diffusion Trajectory Generation

Given the anchors and topology constraints obtained from semantic mapping, trajectory generation is formulated as a conditional multi-agent synthesis problem. An interaction-guided diffusion model is employed to produce trajectories that are both behaviorally realistic and structurally consistent. The diffusion model captures human driving priors, while anchors and lane-level constraints impose geometric structure. Their joint conditioning enables trajectories that align with both interaction semantics and the target closed-track topology.

Let $\mathcal{X} = \{X^0, X^1, \dots, X^N\}$ denote the joint state trajectories of all participating vehicles, and let $\mathcal{U} = \{U^0, U^1, \dots, U^N\}$ denote the corresponding joint control sequence. For each vehicle i , the state and control trajectories are written as $X^i = \{x_t^i\}_{t=0}^T$ and $U^i = \{u_t^i\}_{t=0}^{T-1}$, respectively. The state of vehicle i at time step t is defined as $x_t^i = [p_{x,t}^i, p_{y,t}^i, \theta_t^i, v_t^i]^\top$, where $p_{x,t}^i$ and $p_{y,t}^i$ denote the planar coordinates, θ_t^i is the heading angle, and v_t^i is the speed. The corresponding control input is defined as $u_t^i = [a_t^i, \omega_t^i]^\top$, where a_t^i denotes the longitudinal acceleration and ω_t^i denotes the yaw rate. The state trajectory is obtained by rolling out the discrete vehicle dynamics

$$x_{t+1}^i = f(x_t^i, u_t^i) \quad (4)$$

where $f(\cdot)$ denotes the discrete-time vehicle dynamics model, as defined in our prior work [11]. Accordingly, the joint trajectory is generated as

$$\mathcal{X} = \text{Rollout}(\mathcal{X}_0, \mathcal{U}) \quad (5)$$

where $\mathcal{X}_0$ denotes the initial states of all participating vehicles.

Let $c = (\mathcal{M}^*, \mathcal{A})$ denote the conditional context for trajectory generation, where $\mathcal{A}$ is the set of interaction anchors. The objective is to generate a joint control sequence

$\mathcal{U}$ conditioned on c , such that the resulting state trajectory $\mathcal{X}$ is realistic, dynamically feasible, and consistent with the prescribed interaction structure.

Instead of directly modeling the state trajectory, the diffusion process is applied to the joint control sequence. Let $\mathcal{U}_k$ denote the noisy joint control sequence at diffusion step k , where $\mathcal{U}_0$ is the clean control sequence sampled from the data distribution $q(\mathcal{U}_0)$, and $\mathcal{U}_K$ gradually approaches an isotropic Gaussian distribution after K diffusion steps. The forward diffusion process is defined as

$$q(\mathcal{U}_{1:K} | \mathcal{U}_0) = \prod_{k=1}^K q(\mathcal{U}_k | \mathcal{U}_{k-1}), \quad (6)$$

$$q(\mathcal{U}_k | \mathcal{U}_{k-1}) = \mathcal{N}(\mathcal{U}_k; \sqrt{1 - \beta_k} \mathcal{U}_{k-1}, \beta_k I)$$

where $q(\cdot)$ denotes the forward noising process and $\{\beta_k\}_{k=1}^K$ is a predefined variance schedule controlling the amount of injected Gaussian noise at each diffusion step.

Using the standard reparameterization form, the noisy control sequence at diffusion step k can be written as

$$\mathcal{U}_k = \sqrt{\alpha_k} \mathcal{U}_0 + \sqrt{1 - \alpha_k} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \quad (7)$$

where ϵ is standard Gaussian noise, $\alpha_k = 1 - \beta_k$, and $\bar{\alpha}_k = \prod_{j=1}^k \alpha_j$.

At inference time, the generation process starts from Gaussian noise $\mathcal{U}_K \sim \mathcal{N}(0, I)$ and iteratively applies a learned denoising model to recover a clean joint control sequence. The reverse diffusion process is written as

$$p_\theta(\mathcal{U}_{0:K} | c) = p(\mathcal{U}_K) \prod_{k=1}^K p_\theta(\mathcal{U}_{k-1} | \mathcal{U}_k, c), \quad (8)$$

$$p_\theta(\mathcal{U}_{k-1} | \mathcal{U}_k, c) = \mathcal{N}(\mathcal{U}_{k-1}; \mu_\theta(\mathcal{U}_k, k, c), \Sigma_k)$$

where $p_\theta(\cdot)$ denotes the learned reverse denoising process parameterized by θ , and Σ_k is a preset covariance term following the variance schedule.

The reverse-process mean is parameterized through a noise prediction network $\epsilon_\theta(\mathcal{U}_k, k, c)$. Specifically, the reconstructed clean control sequence is estimated as

$$\hat{\mathcal{U}}_0 = \frac{1}{\sqrt{\bar{\alpha}_K}} (\mathcal{U}_K - \sqrt{1 - \bar{\alpha}_K} \epsilon_\theta(\mathcal{U}_K, K, c)) \quad (9)$$

To further enforce structural consistency during sampling, we introduce a differentiable trajectory-level guidance energy. Since the geometric constraints are imposed on the state trajectory rather than directly on the control sequence, the guidance is evaluated in the trajectory space and propagated back to the control sequence through differentiable rollout dynamics. The guidance objective is evaluated on the reconstructed clean trajectory $\hat{\mathcal{X}}_0$ and is defined as

$$E(\hat{\mathcal{X}}_0 | c) = \lambda_{\text{anch}} E_{\text{anch}}(\hat{\mathcal{X}}_0 | c) + \lambda_{\text{bound}} E_{\text{bound}}(\hat{\mathcal{X}}_0 | c) \quad (10)$$

where λ_{anch} and λ_{bound} are weighting coefficients for anchor alignment and lane-boundary protection, respectively.

Anchor alignment. To preserve the critical interaction structure extracted in the scenario semantic mapping stage, the mapped anchors are organized into vehicle-wise anchor sets. For each participating vehicle i , its anchor set is denoted by

$$A^i = \{(\mathbf{p}_{i,a}, t_{i,a})\}_{a=1}^{N_a^i} \quad (11)$$

where $(\mathbf{p}_{i,a}, t_{i,a})$ denotes the spatial position and corresponding anchor time of the a -th anchor associated with vehicle i , and N_a^i is the number of anchors assigned to vehicle i . Let $\hat{\mathbf{p}}^i(t)$ denote the predicted position of vehicle i on the reconstructed trajectory at time t . The anchor alignment term is defined as

$$E_{\text{anch}}(\hat{\mathcal{X}}_0 | c) = \sum_{i=1}^N \sum_{(\mathbf{p}, t) \in A^i} \|\hat{\mathbf{p}}^i(t) - \mathbf{p}\|_2^2 \quad (12)$$

Lane boundary protection. To prevent generated trajectories from approaching or crossing solid lane boundaries, a differentiable protection term is introduced. Let $\mathbf{b}_t^i$ denote the nearest point on the solid lane boundary to $\hat{\mathbf{p}}^i(t)$, as defined by the map topology $\mathcal{M}^*$. The distance between the trajectory point and the boundary is written as

$$d_t^i = \|\hat{\mathbf{p}}^i(t) - \mathbf{b}_t^i\|_2 \quad (13)$$

Given a preset safety margin d_{safe} , the lane-boundary term is defined as

$$E_{\text{bound}}(\hat{\mathcal{X}}_0 | c) = \sum_{i=1}^N \sum_{t=0}^T \phi(d_{\text{safe}} - d_t^i) \quad (14)$$

At diffusion step k , both the noisy joint control sequence $\mathcal{U}_k$ and the reconstructed clean control sequence $\hat{\mathcal{U}}_0$ are rolled out from the initial state $\mathcal{X}_0$ through the differentiable vehicle dynamics, yielding the corresponding trajectories $\mathcal{X}_k$ and $\hat{\mathcal{X}}_0$, respectively. Since the guidance energy is defined in the trajectory space, its gradient can be back-propagated to the control sequence through the rollout dynamics. Accordingly, the reconstructed clean control sequence is corrected as

$$\tilde{\mathcal{U}}_0 = \hat{\mathcal{U}}_0 - \eta_k \nabla_{\hat{\mathcal{U}}_0} E(\hat{\mathcal{X}}_0 | c) \quad (15)$$

where η_k denotes the guidance scale at diffusion step k .

The corrected control sequence is then used to construct the guided reverse-process mean

$$\tilde{\mu}_\theta(\mathcal{U}_k, k, c) = \frac{\sqrt{\alpha_k} \beta_k}{1 - \alpha_k} \tilde{\mathcal{U}}_0 + \frac{\sqrt{\alpha_k} (1 - \alpha_{k-1})}{1 - \alpha_k} \mathcal{U}_k \quad (16)$$

thereby steering each denoising step toward lower-energy samples while preserving the learned behavioral prior.

3.3.2. Interaction Risk Regulation

During real-world closed-loop execution, the ego vehicle is controlled by the system under test, while BVs are inevitably affected by vehicle dynamics, tracking errors, and

external disturbances. Consequently, predefined trajectories alone cannot reliably maintain the intended conflict timing, interaction evolution, or risk level, making realistic and controllable reproduction of interactive events challenging.

To address this issue, we formulate the interaction between the ego vehicle and the critical BV as a dynamic game and develop an interaction risk regulation mechanism that preserves event semantics while enabling online regulation of interaction intensity. The ego vehicle, serving as the vehicle under test (VUT), is modeled as the *follower*, while the critical BV is modeled as the *leader*. The interaction dynamics are described in Frenet coordinates defined with respect to the corresponding reference path. For each vehicle, the state and control input are defined as $\mathbf{x} = [s, \dot{s}, l, \dot{l}]^\top$ and $\mathbf{u} = [\ddot{s}, \ddot{l}]^\top$, respectively. A simplified discrete-time state-space model is adopted to characterize the vehicle kinematics.

$$\mathbf{x}(k+1) = A \mathbf{x}(k) + B \mathbf{u}(k) \quad (17)$$

where,

$$A = \begin{bmatrix} 1 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & \Delta t \\ 0 & 0 & 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} \frac{\Delta t^2}{2} & 0 \\ \Delta t & 0 \\ 0 & \frac{\Delta t^2}{2} \\ 0 & \Delta t \end{bmatrix} \quad (18)$$

During test-time on the real-world testing, BV is required to remain close to the reference trajectories while inducing adversarial interactions for risk exposure. Accordingly, the *leader*'s interaction cost function is defined as:

$$J_L = \omega_{\text{adv}} J_{\text{adv}} + \omega_{\text{real}} J_{\text{real}} \quad (19)$$

where J_{adv} is the adversarial reward that promotes risk exposure, and J_{real} is the realism reward that encourages feasible execution and adherence to the reference trajectories. The weighting coefficients ω_{adv} and ω_{real} balance adversariality and realism.

Adversarialness. Post-encroachment time (PET) [38] is adopted to quantify interaction risk. Rather than simply minimizing PET, we introduce a target PET value to achieve a desired interaction risk. The adversarial term is defined as the squared deviation between the realized PET and the target PET:

$$J_{\text{adv}} = (\text{PET}(L, F, CP, \gamma) - \tau)^2 = \begin{cases} (t_{L,CP} - t_{F,CP} - \tau)^2, & \gamma = \text{rush}, \\ (t_{F,CP} - t_{L,CP} - \tau)^2, & \gamma = \text{yield}. \end{cases} \quad (20)$$

where L and F denote the leader and follower, respectively, and CP denotes the conflict point. The variables $t_{L,CP}$ and $t_{F,CP}$ represent the arrival times of vehicles L and F at CP , respectively. The variable γ specifies the interaction style, while τ denotes the target PET value.

Realism. Realism is characterized by the deviation from the reference trajectory together with a penalty on excessive control effort. The realism term is defined as

$$J_{\text{real}} = \sum_{k=1}^T \left(\left\| \mathbf{x}_k^L - \mathbf{x}_{\text{ref},k}^L \right\|_Q^2 + \left\| \mathbf{u}_k^L \right\|_R^2 \right) \quad (21)$$

where the first term encourages the BV to remain close to the reference trajectory, while the second term promotes smooth and physically feasible control actions.

The *follower* is modeled to account only for realism, and its cost function is defined as:

$$\begin{aligned} \min_{\mathbf{u}_{k:k+N-1}^F} \quad & J_F = J_{\text{real}} \\ & = \sum_{i=k}^{k+N-1} \left(\left\| \mathbf{x}_i^F - \mathbf{x}_{\text{ref},i}^F \right\|_Q^2 + \left\| \mathbf{u}_i^F \right\|_R^2 \right) \quad (22) \\ \text{s.t.} \quad & \mathbf{x}_{i+1}^F = f(\mathbf{x}_i^F, \mathbf{u}_i^F), \quad \mathbf{u}_i^F \in [a_{\min}, a_{\max}], \\ & s_i^F - s_i^L \geq s_{\text{safe}}, \quad l_i^F - l_i^L \geq l_{\text{safe}} \end{aligned}$$

The *leader's* cost function needs to account for the optimal trajectory of the *follower* in order to calculate the adversarial interactions. The resulting optimization problem is formulated as

$$\begin{aligned} \min_{\mathbf{u}_{k:k+N-1}^L} \quad & J_L = \omega_{\text{real}} \sum_{i=k}^{k+N-1} \left(\left\| \mathbf{x}_i^L - \mathbf{x}_{\text{ref},i}^L \right\|_Q^2 + \left\| \mathbf{u}_i^L \right\|_R^2 \right) \\ & + \omega_{\text{adv}} (\text{PET}(L, F, CP, \gamma) - \tau)^2 \quad (23) \\ \text{s.t.} \quad & \mathbf{x}_{i+1}^L = f(\mathbf{x}_i^L, \mathbf{u}_i^L), \\ & \mathbf{u}_i^L \in [a_{\min}, a_{\max}], \\ & (\mathbf{x}^F, \mathbf{u}^F) \in \arg \min_{\mathbf{u}_{k:k+N-1}^F} J_F(\mathbf{x}^L, \mathbf{x}^F, \mathbf{u}^F) \end{aligned}$$

The above interaction modeling can be formulated as a Stackelberg bilevel optimal control problem, in which the leader optimizes its control sequence in the outer layer, while the follower computes its best-response trajectory in the inner layer conditioned on the leader's strategy. To solve this problem numerically, we adopt a hierarchical nested scheme. For each candidate leader strategy considered in the outer layer, the inner layer solves the follower's optimal control problem to obtain the corresponding response trajectory. Both layers are implemented in CasADi and solved using a nonlinear programming solver. The follower response is then fed back to update the leader's objective and constraints, yielding a risk-adjusted leader trajectory. The overall bilevel optimization problem is formulated as follows.

$$\begin{aligned} (\mathbf{u}^{L*}, \mathbf{x}^{L*}) &= \arg \min_{\mathbf{u}^L} \min_{\mathbf{u}^F \in \Gamma(\mathbf{u}^L)} J_L(\mathbf{x}^L, \mathbf{x}^F, \mathbf{u}^F, \mathbf{u}^L, \omega, \gamma, \tau) \\ \text{s.t.} \quad & \Gamma(\mathbf{u}^L) \\ & = \left\{ \xi \in \Phi^2 : J_F(\mathbf{x}^L, \mathbf{x}^F, \xi, \mathbf{u}^L) \leq J_F(\mathbf{x}^L, \mathbf{x}^F, \mathbf{u}^F, \mathbf{u}^L), \right. \\ & \quad \left. \forall \mathbf{u}^F \in [a_{\min}, a_{\max}] \right\}, \gamma \in \{\text{rush}, \text{yield}\}. \end{aligned}$$

Algorithm 1: Interaction-Guided Steerable Adversarial Testing

Input: Target map topology $\mathcal{M}^*$, initial state $\mathcal{X}_0$, conditional context c , diffusion steps K , optimization horizon H , risk parameter ω , interaction style γ , target interaction risk τ

Output: Reference trajectories $(\mathbf{x}_{\text{ref}}^F, \mathbf{x}_{\text{ref}}^L)$ and executable BV trajectory $\mathbf{x}^{L*}$

```

1 Initialize noisy action sequence  $\mathcal{U}_K \sim \mathcal{N}(0, \mathbf{I})$ ;
2 for  $k \leftarrow K$  to 1 do
3    $\hat{\epsilon}_k \leftarrow \epsilon_\theta(\mathcal{U}_k, k, c)$ ;
4    $\hat{\mathcal{U}}_0 \leftarrow \frac{1}{\sqrt{\bar{\alpha}_k}} \left( \mathcal{U}_k - \sqrt{1 - \bar{\alpha}_k} \hat{\epsilon}_k \right)$ ;
5    $\hat{\mathcal{X}}_0 \leftarrow \text{Rollout}(\mathcal{X}_0, \hat{\mathcal{U}}_0)$ ;
6    $E \leftarrow \lambda_{\text{anch}} E_{\text{anch}}(\hat{\mathcal{X}}_0 | c) + \lambda_{\text{bound}} E_{\text{bound}}(\hat{\mathcal{X}}_0 | c)$ ;
7    $\tilde{\mathcal{U}}_0 \leftarrow \hat{\mathcal{U}}_0 - \eta_k \nabla_{\hat{\mathcal{U}}_0} E$ ;
8    $\tilde{\mu}_\theta(\mathcal{U}_k, k, c) \leftarrow \frac{\sqrt{\bar{\alpha}_{k-1}} \beta_k}{1 - \bar{\alpha}_k} \tilde{\mathcal{U}}_0 + \frac{\sqrt{\bar{\alpha}_k(1 - \bar{\alpha}_{k-1})}}{1 - \bar{\alpha}_k} \mathcal{U}_k$ ;
9    $\mathcal{U}_{k-1} \leftarrow \tilde{\mu}_\theta(\mathcal{U}_k, k, c)$ ;
10  $(\mathbf{x}_{\text{ref}}^F, \mathbf{x}_{\text{ref}}^L) \leftarrow \text{Rollout}(\mathcal{X}_0, \mathcal{U}_0)$ ;
11 while the game testing episode is active do
12   Initialize  $J_F \leftarrow 0, J_L \leftarrow 0$ ;
13   for  $t \leftarrow 0$  to  $H - 1$  do
14      $J_F \leftarrow J_F + \left\| \mathbf{x}_t^F - \mathbf{x}_{\text{ref},t}^F \right\|_Q^2 + \left\| \mathbf{u}_t^F \right\|_R^2$ ;
15      $J_L \leftarrow$ 
16        $J_L + \omega_{\text{real}} \left( \left\| \mathbf{x}_t^L - \mathbf{x}_{\text{ref},t}^L \right\|_Q^2 + \left\| \mathbf{u}_t^L \right\|_R^2 \right) +$ 
17        $\omega_{\text{adv}} (\text{PET}(L, F, CP, \gamma) - \tau)^2$ ;
18    $\mathbf{x}^{L*} \leftarrow$ 
19      $\arg \min_{\mathbf{u}^L} \min_{\mathbf{u}^F \in \Gamma(\mathbf{u}^L)} J_L(\mathbf{x}^L, \mathbf{x}^F, \mathbf{u}^F, \mathbf{u}^L, \omega, \gamma, \tau)$ ;

```

(24)

During closed-track execution, the BV follows the risk-regulated trajectory generated online by the game-theoretic regulator, thereby realizing the intended interaction scenario. The complete procedure is summarized in Algorithm 1.

4. Experiments

To evaluate whether Video2Track satisfies the three core requirements introduced in Section 1, namely *fidelity*, *adversariality*, and *steerability*, we organize the experiments around the following research questions:

- **RQ1:** Can Video2Track faithfully reproduce interaction scenarios from real-world videos?
- **RQ2:** Can Video2Track generate more adversarial interactions to support adversarial stress testing?
- **RQ3:** Can Video2Track generate steerable variants in risk level and interaction style?

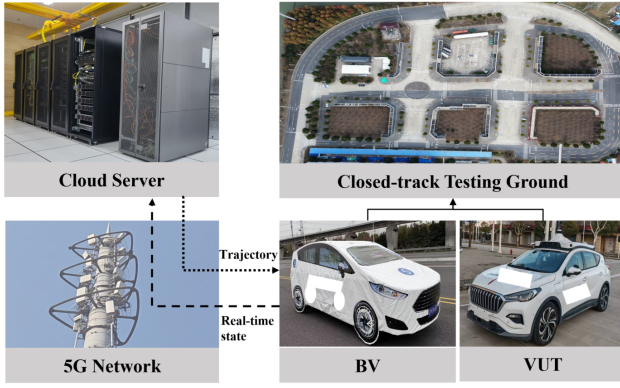


Figure 2: Deployment of Video2Track on the cloud-controlled closed-track testing platform

4.1. Experimental Setup

4.1.1. Platform and Deployment

The proposed Video2Track is deployed and validated on a cloud-controlled testing platform at a university-operated intelligent vehicle proving ground, as illustrated in Fig. 2. All algorithmic modules are executed on a cloud server equipped with an NVIDIA RTX 4090 GPU, where BV trajectories are generated and updated online. The generated trajectories are transmitted to BVs through a dedicated 5G network via the MQTT protocol. Each BV is equipped with a high-precision localization module and an onboard control execution unit. A longitudinal PID controller and a lateral Stanley controller are adopted for closed-loop trajectory tracking. Meanwhile, vehicle states are continuously fed back to the cloud server for online state update and adversarial regulation.

4.1.2. Compared methods

Predefined Scenario: This baseline represents conventional standardized scenario-based testing, where traffic participants follow predefined trajectories without online adaptation to the VUT. It serves as a non-interactive reference.

TroubleMaker [10]: This method extends predefined scenario testing by introducing an online interaction regulation mechanism that actively induces conflicts with the VUT. It serves as a representative adversarial testing baseline.

Diffusion-based Scenario: This baseline adopts the diffusion-based trajectory generation method in [16] to generate BV trajectories, which are then adapted to the closed-track execution interface for deployment.

Video2Track (ours): Our framework combines scenario semantic mapping and steerable adversarial testing.

4.1.3. Metrics

Evaluation of the generated testing scenarios is conducted from three aspects: interaction fidelity, adversariality, and steerability.

Interaction Fidelity: Qualitative assessment focuses on the consistency of interaction structure, temporal passing order, and conflict evolution between the source video and the reproduced closed-track execution. Quantitatively, fidelity is

evaluated by the minimum PET error

$$\min PET_{\text{error}} = |PET_{\text{track}}^{\min} - PET_{\text{video}}| \quad (25)$$

where PET_{video} denotes the PET label associated with the source video, and $PET_{\text{track}}^{\min}$ denotes the minimum PET measured from the reproduced closed-track scenario. A smaller $\min PET_{\text{error}}$ indicates closer agreement in interaction criticality between the reproduced scenario and the source video.

Adversariality: Adversariality is quantified by the mean minimum PET over repeated closed-track executions, denoted as $\text{mean} PET_{\text{track}}^{\min}$. A smaller $\text{mean} PET_{\text{track}}^{\min}$ indicates stronger adversarial pressure.

Steerability: It is examined in terms of risk-level controllability and interaction-style controllability. Risk-level controllability is quantified by the error between the target PET and the achieved $PET_{\text{track}}^{\min}$ in repeated closed-track executions, where a smaller error indicates more accurate control of the intended interaction risk. Interaction-style controllability is evaluated qualitatively by comparing generated scenarios under different style settings, such as *rush* and *yield*.

4.1.4. Experiment Design

Experimental evaluation is organized around two representative interaction scenario categories, namely unprotected left-turn and cut-in, both of which are typical conflict scenarios in ADS evaluation. These scenarios involve common urban interactions with clear conflict structures and moderate vehicle speeds, making them suitable for safe and repeatable execution in closed-track environments. For each category, multiple closed-track experiments are conducted to assess the fidelity of scenario migration, the achievable adversariality, and the steerability of risk level and interaction style.

4.2. Results Analysis

4.2.1. RQ1: Can Video2Track faithfully reproduce interaction scenarios from real-world videos?

First, we qualitatively compare the critical frames extracted from the source videos with the corresponding closed-track testing results. Fig. 3 and Fig. 4 present representative comparisons for the cut-in and unprotected left-turn scenarios, respectively. In the cut-in scenario, the VUT proceeds straight in the rightmost lane while a BV from the adjacent left lane cuts into the VUT's lane. The VUT then performs a yield maneuver, after which the BV completes the merge. In the unprotected left-turn scenario, the VUT turns left at the intersection while interacting with an oncoming BV from the opposite through lane; the BV yields and the VUT passes through the conflict area. In both scenarios, the reproduced closed-track testing results preserve the original interaction type and the behaviors of both vehicles, indicating that the proposed method can faithfully transfer real-world interaction events into closed-track testing.

We further quantitatively evaluate scenario fidelity using PET error. Specifically, as defined in Eq. 25, the minimum

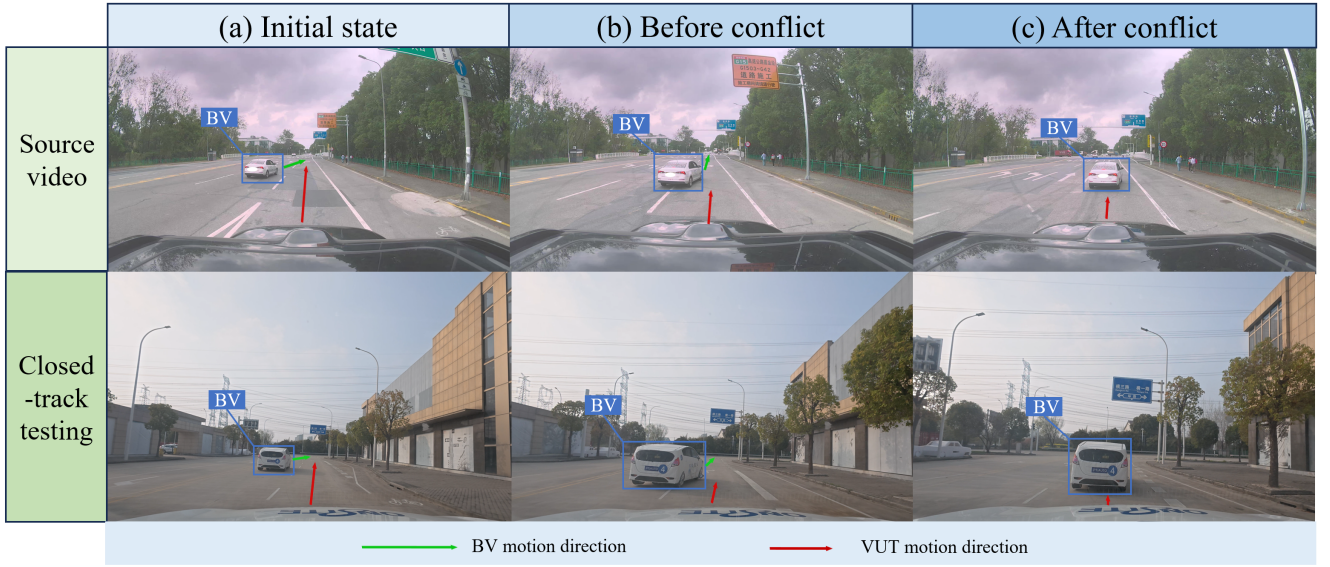


Figure 3: Comparison between the source video and closed-track testing in the cut-in scenario

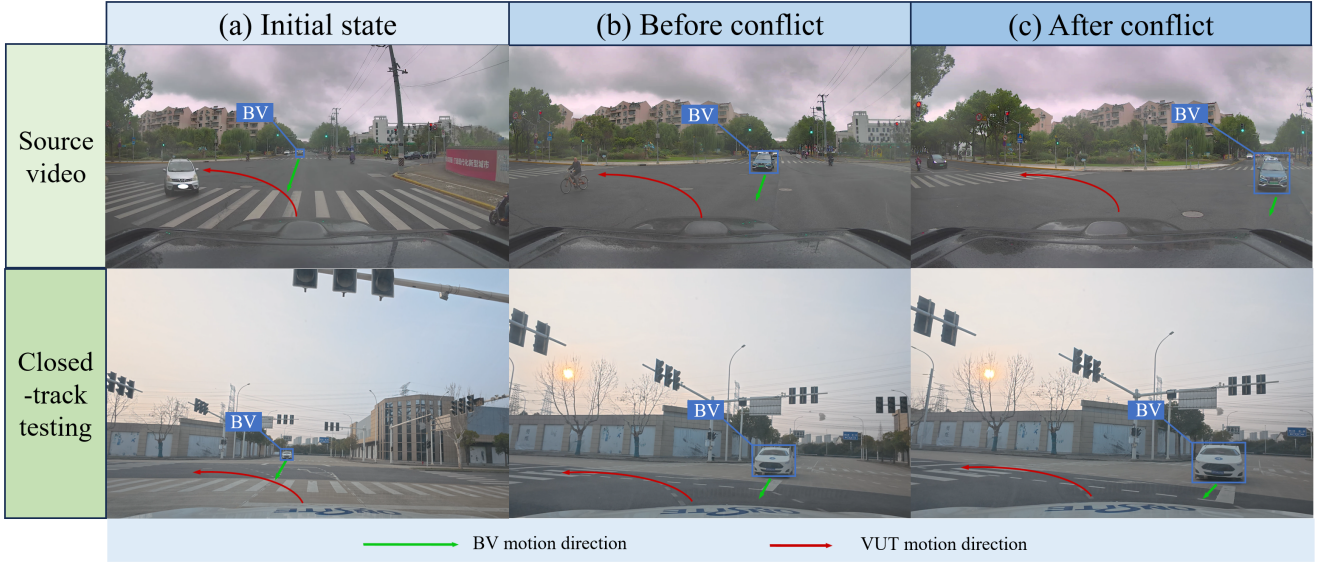


Figure 4: Comparison between the source video and closed-track testing in the unprotected left-turn scenario

PET achieved in the closed-track executions is compared with the PET annotated in the source video. As shown in Fig. 5 and Fig. 6, the relative PET error is 7.0% for the cut-in scenario, where the video PET is 4.70s and the measured minPET in the closed-track test is 5.03s. For the unprotected left-turn scenario, the relative PET error is 9.7%, where the video PET is 2.90s and the measured minPET is 3.18s. These small deviations indicate strong consistency in interaction intensity between the reproduced tests and the source videos. Together with the qualitative results, this demonstrates that Video2Track achieves high interaction fidelity in scenario reproduction.

Table 1

Comparison of adversarial stress testing capability across different methods.

Method	$meanPET_{track}^{min} (s)$	
	Cut-in	Unprotected Left-turn
Predefined	2.5	2.2
TroubleMaker	1.6	1.7
Diffusion	1.7	1.9
Video2Track(our)	1.3	1.5

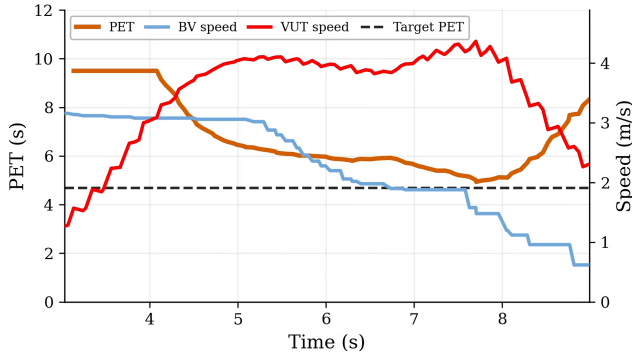


Figure 5: The cut-in PET of closed-track testing

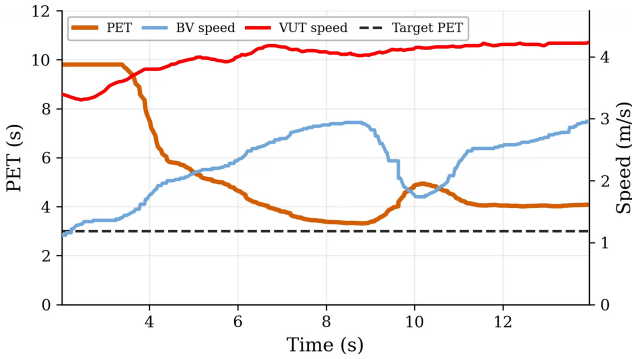


Figure 6: The unprotected left-turn PET of closed-track testing

4.2.2. RQ2: Can Video2Track support adversarial stress testing ?

Table 1 compares the adversariality of different methods in the cut-in and unprotected left-turn scenarios, measured by $meanPET_{track}^{min}$. The predefined scenario yields the largest $meanPET_{track}^{min}$ values in both scenarios, indicating relatively conservative interactions due to the lack of active adversarial regulation. TroubleMaker reduces $meanPET_{track}^{min}$ in both scenarios, suggesting that it can increase interaction criticality to a certain extent. The diffusion-based method also achieves lower $meanPET_{track}^{min}$ than the predefined scenario, but still underperforms the proposed method, partly because the generated trajectories may exhibit relatively large curvature, which can introduce tracking deviations for the BV during closed-track execution. In contrast, Video2Track achieves the lowest $meanPET_{track}^{min}$ in both the cut-in and unprotected left-turn scenarios, with values of 1.3s and 1.5s, respectively. These results demonstrate that Video2Track can generate more critical interactions than the baseline methods while preserving the same interaction type, thereby enabling stronger adversarial risk and more challenging stress testing in closed-track testing.

4.2.3. RQ3: Can Video2Track generate steerable variants in risk level and interaction style ?

We evaluate the proposed Video2Track framework from two aspects: steerable risk generation and interaction-style generalization.

Table 2

Steerable interaction risk test results.

Target PET (s)	$meanPET_{track}^{min}$ (s)	$meanPET_{error}$ (%)
2.0	1.81	9.5
3.0	3.13	4.3
4.0	4.21	5.3
5.0	5.37	7.4

First, we examine whether Video2Track can generate scenarios with steerable interaction risk levels. Using the unprotected left-turn scenario as an example, we set the target PET to 2.0s, 3.0s, 4.0s, and 5.0s, respectively, and compare them with the mean minimum PET values achieved in the generated testing scenarios. As shown in Fig. 7 and Table 2, the achieved $meanPET_{track}^{min}$ increases consistently with the prescribed target, indicating that the interaction risk of the generated scenarios can be adjusted in a controllable manner. Although slight deviations are observed, the relative error remains below 10% in all cases. These results demonstrate that Video2Track enables controllable generation of scenarios with different interaction risk levels while preserving the semantic consistency of the original interaction.

Second, we evaluate whether Video2Track can achieve controllable interaction-style generalization. Starting from the same unprotected left-turn scenario, we modify the BV interaction style from yielding to rushing while keeping the interaction type unchanged. As shown in Fig. 8a and Fig. 8b, both generated scenarios preserve the same underlying unprotected left-turn interaction, while exhibiting clearly different motion patterns and behavioral tendencies. In particular, the BV behavior is successfully transformed from a conservative yielding style to a more aggressive rushing style. This demonstrates that the proposed framework can generalize a single real-world interaction instance into multiple style-controllable variants without altering the original interaction semantics. For clearer visualization, the Supplementary Material provides two execution videos, *unprotected left-turn_style.mp4* and *cut-in_style.mp4*, covering four representative cases: unprotected left-turn and cut-in scenarios with BV yielding and rushing behaviors.

Overall, the above results verify that Video2Track can not only preserve the semantic structure of the source scenario, but also generate steerable scenario variants in both prescribed risk level and interaction style.

5. Conclusion

This paper presented Video2Track, a framework for transferring real-world driving interaction videos into steerable adversarial closed-track testing scenarios for ADS. The framework consists of two main components: a scenario semantic mapping module, which combines VLM-based semantic extraction with topology-library retrieval for semantic grounding, and a dynamic interactive testing

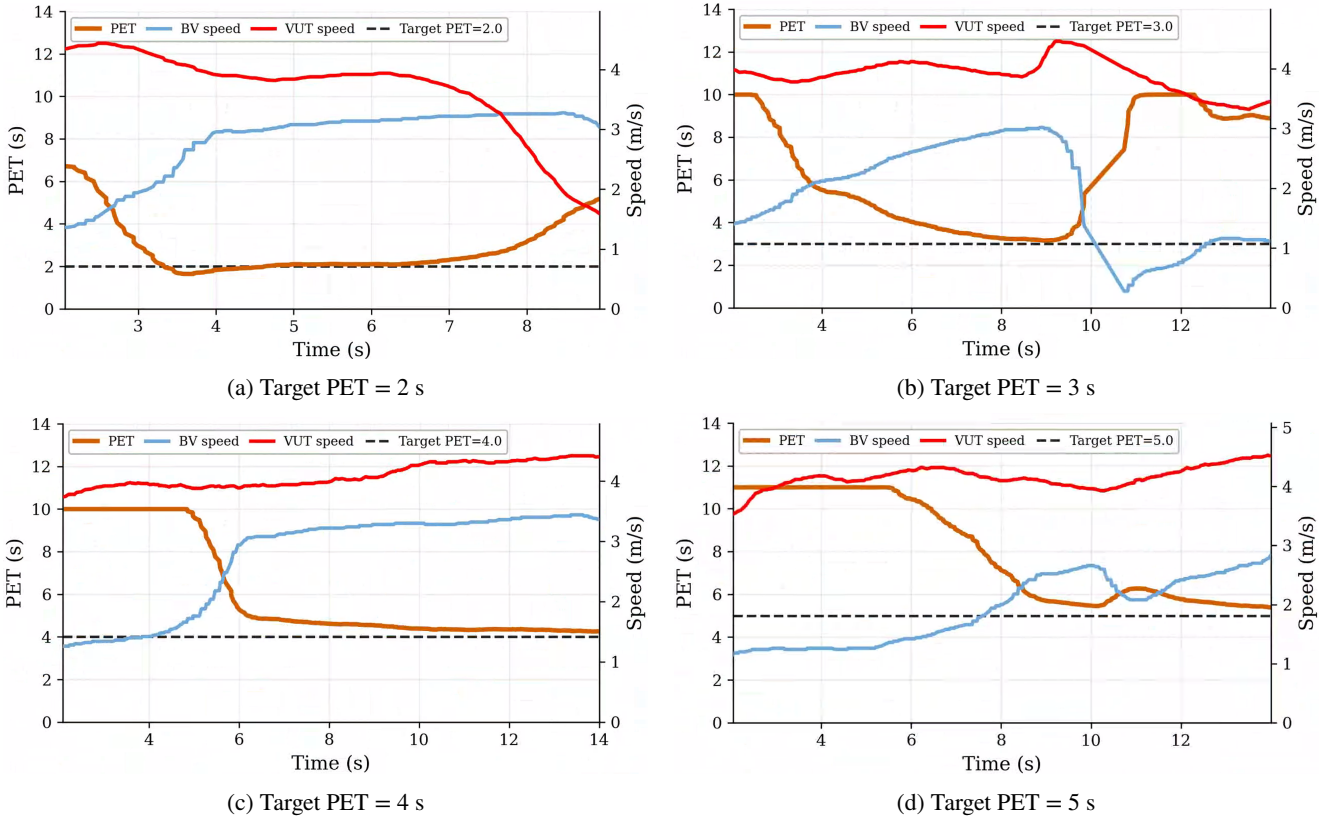


Figure 7: Steerable interaction risk test results.

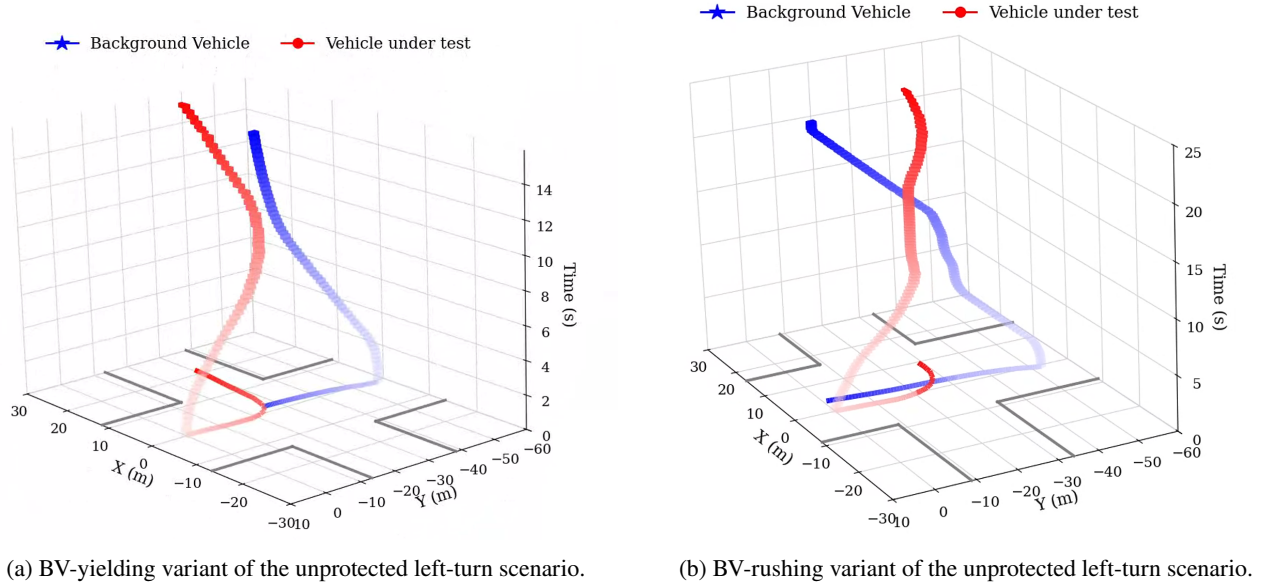


Figure 8: Interaction style generalization in the unprotected left-turn scenario.

module, which integrates interaction-guided diffusion trajectory generation with Stackelberg-based interaction risk regulation. Closed-track experiments on a cloud-controlled testing platform demonstrate that Video2Track can faithfully reproduce the interaction semantics of real-world videos, while enabling continuous regulation of interaction risk and controllable variation in interaction style. Compared

with representative baseline methods, Video2Track achieves stronger interaction adversariality while maintaining semantic fidelity to the source scenarios. These results demonstrate that Video2Track satisfies three key requirements for closed-track testing, including fidelity, adversariality, and steerability.

Future work will focus on extending the framework from single interaction events to continuous scenario testing, so as to support the reproduction and regulation of long-horizon, multi-stage interactions in more realistic driving processes.

A. Prompt Flow for Scenario Abstraction

This appendix presents the prompt design in Fig. A1 used for scenario semantic abstraction. The prompt guides the VLM to extract structured semantics from driving videos, including road topology, agent states, and interaction events.

B. Closed-track Map Topology Library

This appendix presents an example entry from the closed-track map topology library. As shown in Fig. A2, each topology block encodes structured road attributes, anchor nodes, maneuver paths, and conflict anchors to support semantic grounding to the target closed-track topology.

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Prompt Flow for Scenario Abstraction

<Role Setting>

You are a traffic scenario interaction semantic analyzer for closed-track testing. Your task is to extract structured scenario semantics from real-world interaction videos. The output should follow a predefined structured format.

<Task>

Given a sequence of driving video frames from the ego vehicle's perspective, along with the corresponding textual label, analyze the interaction event and extract structured scenario semantics. You should follow a step-by-step reasoning process.

<Reasoning>

Step 1: Identify the road type and lane structure(e.g., intersection-2lane, straight-2lane).

Step 2: Infer the motion behavior of the ego vehicle under the identified road topology in Step 1, including its current lane and maneuver intention (e.g., going straight, turning left, turning right, lane changing, merging).

Step 3: Detect the main background vehicles that interact with the ego vehicle and describe their driving behaviors.

Step 4: Infer the interaction event and interaction type between the ego vehicle and interacting background vehicle based on their trajectory relationships (e.g., Head-On, Crossing Path, Merging Path, Following).

Step 5: Determine the interaction style of each interacting background vehicle with respect to the ego vehicle (e.g., Aggressive, Yielding) based on observed behavior patterns.

<Output>

Provide the output as structured scenario semantics with the following fields:

- Road Topology: road type and lane structure
- Ego Vehicle State: driving behavior, current lane, maneuver intention, and speed range
- Background Vehicle States: driving behaviors, interaction types, and interaction styles
- Interaction Events: interaction relationships between the ego vehicle and background vehicles
- Event Description: a concise natural language summary of the scenario

Example:

Road Topology: Intersection-4lane

Ego Vehicle State: the ego vehicle is in the left-turn lane before the stop line and intends to turn left

Background Vehicle States: the agent is in the opposite through lane, drives straight, and slows down before the conflict point.

Interaction Events: Crossing Path, Yielding

Event Description: The ego vehicle prepares to turn left at a four-lane intersection while an oncoming vehicle proceeds straight and yields near the conflict point.

Figure A1: Prompt flow for scenario abstraction.

Closed-track Map Topology Library

Intersection-3 Inbound Lanes and 2 Outbound Lanes

Topology Description

This block represents a typical urban four-arm intersection, with inbound and outbound approaches in the north, east, south, and west directions. Each inbound approach consists of three functional lanes for left-turn, through, and right-turn movements, while each outbound approach contains two departure lanes. The topology supports multiple maneuvers, including through driving, left turns, and right turns, and is suitable for testing interaction scenarios such as unprotected left turns and unsignalized traffic movements.

Approach and Direction Convention

The four approaches are denoted clockwise as North (N), East (E), South (S), and West (W). For each approach, inbound lanes are denoted by "in" and outbound lanes by "out". Lane indices start from 0 at the lane closest to the road centerline and increase outward.

Key Nodes

- S_in_0: left-turn lane of the south inbound approach, coordinates \$(xx, xx)\$
- S_in_1: through lane of the south inbound approach, coordinates \$(xx, xx)\$
- S_in_2: right-turn lane of the south inbound approach, coordinates \$(xx, xx)\$
- S_out_0: left outbound lane of the south approach, coordinates \$(xx, xx)\$
- S_out_1: right outbound lane of the south approach, coordinates \$(xx, xx)\$
- Conflict_0: associated with the crossing paths $\langle S_in_0 \rightarrow W_out_0 \rangle$ and $\langle N_in_1 \rightarrow S_out_1 \rangle$, with coordinates \$(xx, xx)\$.
- ...

Representative Maneuver Paths

- Left turn: $S_in_0 \rightarrow \langle W_out_0, W_out_1 \rangle$
- Through movement: $S_in_1 \rightarrow \langle N_out_0, N_out_1 \rangle$
- Right turn: $S_in_2 \rightarrow \langle E_out_0, E_out_1 \rangle$

Typical QA Example

Q: How is an unprotected left turn configured in this block?

A: Select the leftmost inbound lane of the south approach (S_in_0) as the starting point and either W_out_0 or W_out_1 on the west outbound approach as the destination; this corresponds to an unprotected left-turn maneuver.

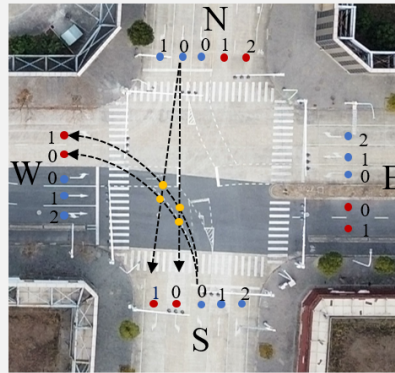


Figure: W, N, S, and E denote the four approach directions. Blue points represent inbound anchors, red points represent outbound anchors, yellow points represent conflict anchors, and dashed lines indicate possible maneuver paths.

Figure A2: Example of an encoded topology block in the closed-track map topology library.